

Research project

Making newspaper layouts dynamic : Study of a new combinatorial/geometric packing problems

1 Context

The digital era is completely transforming the newspaper industry, offering new user experience for all readers. However, to be successful, newspaper designers stand before a hard **design challenge** [IAN04, HKRAA11]: translating the **design and aesthetics** from the printed edition [GG17, Ges18] and the **functionality** from the online edition to create the e-newspaper of the future, making a synthesis based on **usability, reading comfort, engagement** [HKRA11, Mor16, NV18]. In this spirit, our project aims at developing a **novel inclusive digital news reading experience** that will benefit **all readers: you, me**, but also **low vision people** for who newspapers are a way to be part of a well-evolved society.

Currently, one way to read the news in digital form is through **the online edition, available** on a variety of screen sizes thanks to **responsive design**. This form offers several advantages related to technical features (e.g., hyperlinks, the possibility of searching, tuning parameters of the reading display) and to the content itself (e.g., continuous updates). One major flaw is the lack of local design factors [HW05, DB18] and global layout compared to the print edition [IAN04]: Websites mostly present news in a linear fashion, thus missing the editorial organization and aesthetics and resulting in a linear article-reading mode [Ajita15,NV18].

For these reasons, many readers often prefer to read **the electronic version of the print edition** on digital devices (e.g., via kiosk press apps) because this form is closer to the original reading experience (e.g., familiarity in design, clear overview of the content, news valuation). However, its strength, that is its aesthetics serving a well-defined storytelling, is also its weakness: newspaper design is static and changing it requires dedicated content management systems (CMS) and newspaper designers expertise.

2 Objectives

Our motivation is to bridge the quality gap between online news and printed newspaper. In this PhD, we want to develop novel computational approaches for design production that could be used to offer a high quality news reading experience, combining the readability and overview from the printed newspaper with the possibilities of online media such as constant updates, interactivity and personalization. To do so, one central question that we want to explore is **how to make newspaper design dynamic ?**

From a methodological perspective, our goal is to go beyond state-of-the-art layouting methods [DS94,SH09,SA10,GRP+02], which generally produce simple page layouts and do not consider the problem of accessibility. The approach that we envision raises a **new kind of combinatorial/geometric packing problem**. Packing problems are a class of optimization problems in mathematics that involve attempting to pack objects together into one or several containers respecting some constraints, and optimizing some functions [Lodi2002, LMM02,

Vaz01]. In our case, objects correspond to newspaper pieces (e.g., heading, articles, images) and the goal is to place them optimally within a space defined by the page. This work will requires the expertise of D. Mazauric in algorithmics and optimization [BM16, CHM12, CHM+18, CMN16].

Given that this packing problem is considered in the context of newspaper layouting, all the expertise from the company Demain un Autre Jour will be crucial. Thanks to Bruno Génuit and collaborators, our packing problem will not only be a geometrical problem, but it will also include all constraints from newspaper production industry.

Three applications will be considered:

- **Application 1: Layout exploration and optimization.** Here the focus is about constraints satisfaction. Given a newspaper layout with a given content, the goal is to explore all possible alternative layouts enabling designers to select a layout with a better aesthetic score.
- **Application 2: Design-preserving magnification (see Figure 1).** Given a newspaper layout with a given content, the goal is to define a new process to magnify the content dynamically while keeping preserving the most essential information at each scale. Here

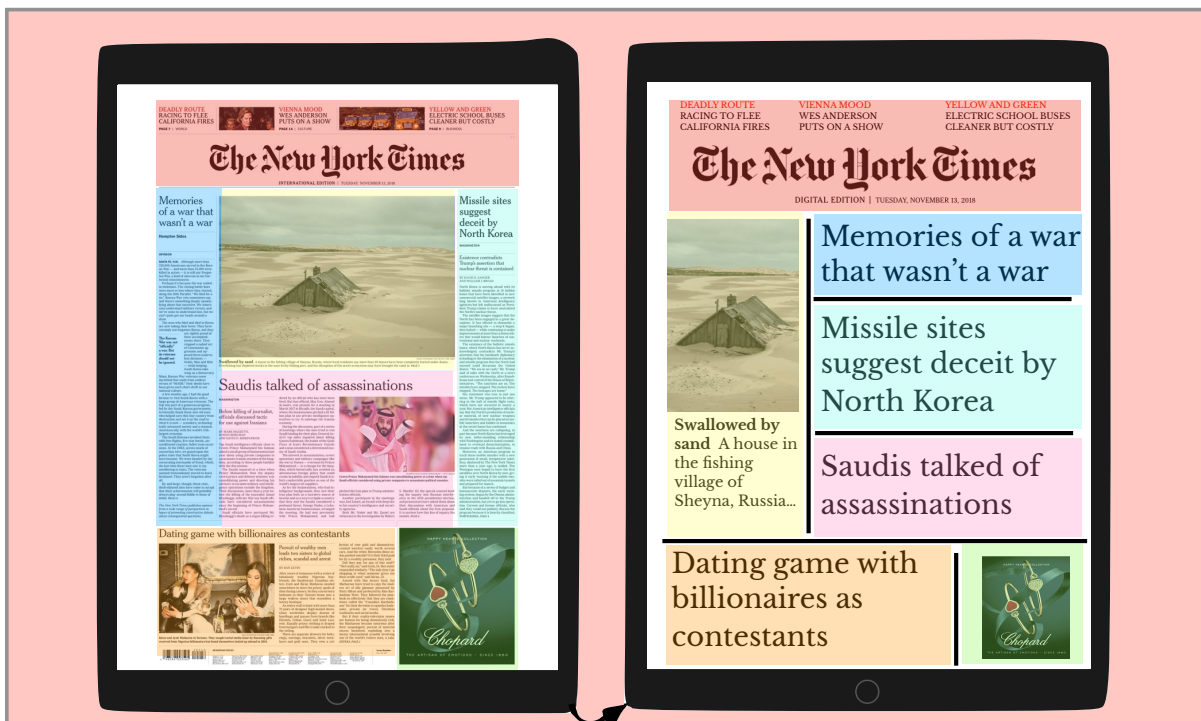


Figure 1: Illustration of design preserving magnification (Application 2). On the **left**, we show how an original newsprint front cover fit into an e-Reader or tablet: text is hardly readable due to the scale. On the **right**, we show how we propose to present the same page (**here, manually built layout**) given a magnification factor. Our approach aims at producing **automatically** a complete redesign of the page according to newspaper design rules, which also fit user needs in terms of readability. **Our solution tackle the local/global navigation problem** encountered by low vision readers who need to magnify to read. This redesign can be seen as a packing problem looking for the best positions and shapes of the blocks of text. A hierarchical description of the page will also be used to decide which elements to preserve or cut, following priority rules. Note that image manipulations (e.g., content-aware cropping) will also be part of the final solution we envision.

the focus is about accessibility: this application targets accessibility for people with vision impairment and more generally any situation where font size is too small (i.e., low visual acuity or reading on small screens). This application will be central in the Ph.D. and benefit from the expertise of Pierre Kornprobst in the domain of low vision and reading behavior.

- **Application 3:** Layouting from scratch / personalized : Given some material, as a further step, we shall investigate our tools and methods to design page layout. Here the focus is about dynamic design: we will need to focus on performance optimisation to produce new layouts with low computational cost. This goal is of particular interest for personalized layout, thanks to Artificial Intelligence in order to learn features most suited for a specific class of readers. It would also offer the possibility to dynamically change parts of the content of a newspaper, while preserving its global aesthetics.

3 Organization of research work

Starting from a preliminary study, we have defined a new kind of packing problem where each block correspond to newspaper elements (e.g., articles, images, advertisement) that should fit in a given space (page surface). Each block could have different shapes. Their positioning should also respect three types of constraints: (1) abstract constraints describing high-level relationship between two components, (2) spatial rules coming from, e.g., priority assigned to some components, (3) aesthetic constraints acting at the page level to provide a pleasant visual experience for the reader.

This packing problem will be the heart and starting point of the Ph.D.. After studying its complexity (e.g., NP-hardness, APX-hardness), the candidate will develop new **algorithmic solutions** to solve the problem in practice, given the high number of variables (around 10^5) and the need to have a method with a low computational cost. The next challenge will be to incorporate progressively **constraints** coming from **newspaper design and aesthetics** [GautierGautier2017,StreckerAlbayrak2002].

Application 1 will provide a first case study. One interesting avenue will be to play with the constraints to observe the changes in the layouts produced. We will compare the results of our approach with the templates defined by Demain un Autre Jour in their Content Management System (Melody software).

Application 2 will require to consider a new type of constraint, i.e., **the visual capabilities of the readers**. We want to produce a new layout **from an original layout**, which will be more **accessible**. The motivation is that, even if intuitive touch screen gestures for **magnification** help the reader with the complex and small print layout of news media (e.g., pinch-to-zoom), the major drawback is the difficulty to **navigate** at both **global** (e.g., between articles, sections, columns, media content) and **local** (e.g., line to line) levels, which impacts reading comfort and **accessibility** importantly. Our ambition is to design a method that automatically generates newspaper layouts with high readability, keeping storytelling unaltered. To do so, we foresee an original approach that we call **design-preserving magnification** (Fig. 2), to solve the local/global navigation problem.

In a nutshell, our idea consists in keeping the global view always self-consistent, even with magnified content. However, of course, one cannot magnify all the content and still fit the same display surface. Thus, this process has two consequences: **(1)** One has to cut some elements from the global view, and the stronger the magnification, the more content has to be cut. To choose which content to cut, we will exploit the hierarchy of the news content (e.g., pages → articles → headline → text → paragraphs) to keep essential components as long as possible. Note that all content will still be available when switching to a local view (e.g., by selecting one article to read it). **(2)** When magnified, the content may not fit well the original layout in terms of aesthetics and design. This is why new layouts should be found to optimize visual aspect while respecting as much as possible original editorial choices.

Two types of evaluations of the resulting designs and user's experience are planned. The first will be done with normally sighted subjects through questionnaires and behavioral analysis (e.g., eye tracking study as in [HW05]), with possible collaboration with UCA partners from Académie 5/CoCoLab platform. The second will be done with low vision subjects, in collaboration with the Centre d'ophtalmologie Monticelli Paradis (Prof. F. Matonti, Marseille) and service d'ophtalmologie du CHU Pasteur (Prof. S. Baillif, Nice), already partner of Biovison Lab.

Application 3 will require to consider an additional problem which is to handle content length and type to be displayed in a page, since we start here from a set of blocks not a priori arranged in an initial layout as this is the case in the two former applications. The performance of the approach, which will be considered at each stage of the Ph.D., will be particularly scrutinised in this context.

4 Expected impact

Given the scientific objectives described above, we expect three kinds of impact:

- **Scientific impact: Bringing an efficient solution to layout production will be a significant breakthrough.** Although several approaches have been developed for layout production, the notion of computational cost is almost nether commented in the literature. Hardly no comparisons or performance analysis are provided in the papers and results are usually limited to a few layouts produced. In this project, we want to develop a rigorous protocole for the evaluation of performance, with a particular focus on computational cost.
- **Societal impact: Ensuring equal access to digital content is essential.** Unfortunately, we are not equal concerning the accessibility to reading material. In particular, newspapers pose a unique challenge to **low vision people** due to their unpredictable layout, timely nature, dynamic content, and condensed formatting [HJ10, LB11, WCK20], making it a burden for news publishers to create accessible versions. Yet news reading has become an essential activity in modern society as a form of social connection, entertainment, and learning [KC13, KLB18]. With an estimated 180 million people suffering from low vision and rapidly rising [BFB+17], there is a strong call for accessible news reading technologies.
- **Economic impact:** By measuring specific aesthetic and geometric criterias, we intend to characterise newspapers' pages to turn a visual layout into a page signature. We will be then able to translate the visual identity of a publisher into a logical matrix. Eventually this deep understanding of the overall look and feel will lead to automatic layouts generation in

full compliance with the visual identity of the publisher. That will result in heavy production costs and delays reduction for Newspapers editors. On a secondary note, ability to score pages on an aesthetic plan could help people monitoring how close or how far their layout is from the graphic ruleset.

5 Bibliographic references

Note: Bibliography combines references from:

- Newspaper design and user experience studies on newspaper reading
- Algorithmics and combinatorial optimization
- Low vision

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